

Artifact Removal Quality Assessment (ARQA): A Recording-Level Audit of EEG Artifact Removal Using Internal Preservation Controls

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Artifact Removal Quality Assessment (ARQA): A Recording-Level Audit of EEG Artifact Removal Using Internal Preservation Controls

Methodological Framework and Proof of Concept with GEDAI

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Abstract

Artifact-removal procedures are commonly judged by reductions in visible contamination or by diagnostics internal to the algorithm, neither of which guarantees preservation of physiological EEG. We present Artifact Removal Quality Assessment (ARQA), a method-agnostic post hoc audit that uses expert-marked artifact-free segments of the original recording as internal negative controls. Before comparison, ARQA harmonizes the channel set, units, polarity, reference, sampling rate,

temporal alignment, and duration. It then reports three distinct domains: preservation in clean windows, artifact suppression in contaminated windows, and the coverage available to estimate each domain. Preservation is quantified using waveform correlation, normalized difference variance, amplitude-scaling ratio, and bandwise spectral change; suppression is evaluated using an artifact-specific marker. We applied ARQA to GEDAI v1.5 and v1.7 outputs from two de-identified, 19-channel, eyes-closed resting-state EEG recordings with contrasting artifact loads. In Recording A, 35.7% of the 2-s windows were available as clean controls. Within those windows, the mean correlation was 0.444 and the normalized difference variance was 82%; across the full recording, delta-band amplitude fell by 91.9%, but alpha-band amplitude also fell by 77.5%, indicating marked nonselective attenuation. In Recording B, 98.6% of the windows were clean controls, the correlation was 0.9973, the normalized difference variance was 0.64%, and clean-window band changes remained within ± 0.6 dB. Only 1.4% of the windows in Recording B were contaminated; thus, suppression efficacy was insufficiently challenged despite excellent preservation. Under GEDAI v1.7, Recording A showed markedly improved preservation (r_{clean} 0.718, alpha attenuation at O1 reduced from -10.1 to -0.4 dB), while Recording B remained excellent with a small frontal regression; both versions produced identical internal SENS AI scores, confirming that external audit adds information beyond internal diagnostics. These cases show why recording-level quality cannot be inferred from algorithm identity or internal scores alone. ARQA is intended as an auditable profile rather than a universal ranking and complements rather than replaces established cleaning-quality metrics.

Keywords: EEG artifact removal; quality control; signal preservation; internal negative control; qEEG; GEDAI; reference harmonization

1 Introduction

Artifacts from eye movements, cranial and facial muscles, head motion, cardiac activity, unstable electrodes, and environmental interference can distort clinical and research EEG. A broad range of countermeasures is available, including epoch rejection, regression, independent component analysis, artifact subspace reconstruction, wavelet methods, multichannel Wiener filtering, and model-informed spatial decompositions (Jiang et al., 2019; Delorme and Makeig, 2004; Chang et al., 2020). The practical question is therefore no longer whether EEG can be cleaned, but whether a particular operation improved a particular recording without removing the neural information that the analysis was intended to preserve.

This distinction is especially important in quantitative EEG (qEEG), in which amplitude and spectral measures are compared with normative models (Bosch-Bayard et al., 2026). A method can produce a visually smooth trace while attenuating physiological alpha, theta, or beta activity. Conversely, a conservative method can preserve the signal while leaving the principal artifact nearly unchanged. Neither visual appearance nor the amount of removed variance is sufficient by itself. Benchmarking studies have accordingly emphasized paired measures of signal preservation and artifact reduction, including signal-to-error and artifact-to-residue ratios, blink-amplitude ratios, and downstream physiological effects (Bailey et al., 2023a, 2023b; Robbins et al., 2020; Hill et al., 2024).

Recent work reinforces the same concern. Bailey et al. (2025) showed that targeting correction to artifact periods and frequencies can preserve more neural signal than applying component subtraction globally. Nyanney et al. (2026) used expert-annotated artifact-free windows to evaluate preservation

using correlation, root-mean-square error, and peak signal-to-noise ratio in a detection-guided clinical EEG pipeline. These studies establish that clean intervals are informative controls. ARQA does not claim that this underlying idea is unprecedented. Its proposed contribution is narrower and operational: a method-agnostic, post hoc, recording-level audit that can be applied to paired original and cleaned outputs, explicitly separates preservation from suppression, and requires pre-comparison signal harmonization.

The present paper defines the ARQA framework and illustrates it using two real recordings processed with the same GEDAI configuration. GEDAI was selected for this proof of concept because it is a state-of-art unsupervised, leadfield-informed approach which provides a technically demanding and clinically relevant test of the ARQA framework. The method uses generalized eigendecomposition to identify activity that diverges from a modelled cortical signal subspace and is designed to process heavily contaminated recordings without requiring clean reference data or manual component classification (Ros et al., 2025). These characteristics make GEDAI especially suitable for demonstrating the need for an independent, recording-level audit: its model-specific diagnostics indicate how well the estimated signal and noise subspaces are separated, while ARQA evaluates the complementary question of whether the processed output preserves physiological EEG within independently defined clean intervals. Applying the same documented GEDAI configuration to recordings with contrasting artifact loads also allowed us to examine whether ARQA could identify recording-dependent differences in preservation and suppression. GEDAI therefore provides a reproducible test case for the framework, with broader comparative evaluation across cleaning methods reserved for the next planned validation phase. The two examples are not presented as a population-level validation or as a verdict on GEDAI. Instead, they serve as deliberately contrasting stress tests of the audit itself.

2 The ARQA framework

2.1 Scope and core principle

Let X denote the original multichannel EEG and Y the output of an arbitrary cleaning procedure M . ARQA asks two separate questions. First, preservation: how much did M change segments that did not require cleaning? Second, efficacy: how much did M reduce the targeted artifact where that artifact was present? A third domain, evaluation coverage, records how much clean and contaminated material was available to estimate the first two domains.

For the present proof-of-concept, an EEG specialist marked windows suitable for qEEG directly in the original recordings before ARQA and before inspection of the GEDAI outputs. These windows are internal negative controls for collateral modification. They are not ground truth for the unknown neural signal hidden beneath an artifact, and they do not prove that contaminated windows were correctly reconstructed. This distinction is central to the interpretation of ARQA.

$$C = \{k : E_k = \text{clean}\}, \quad A = \{1, \dots, K\} \setminus C$$

Here E_k is the expert label assigned to window k , C is the clean-control set, and A is its contaminated complement. The labeling rule must be fixed from X alone and must not depend on Y or on diagnostics produced by the algorithm being evaluated. Otherwise, the audit becomes circular.

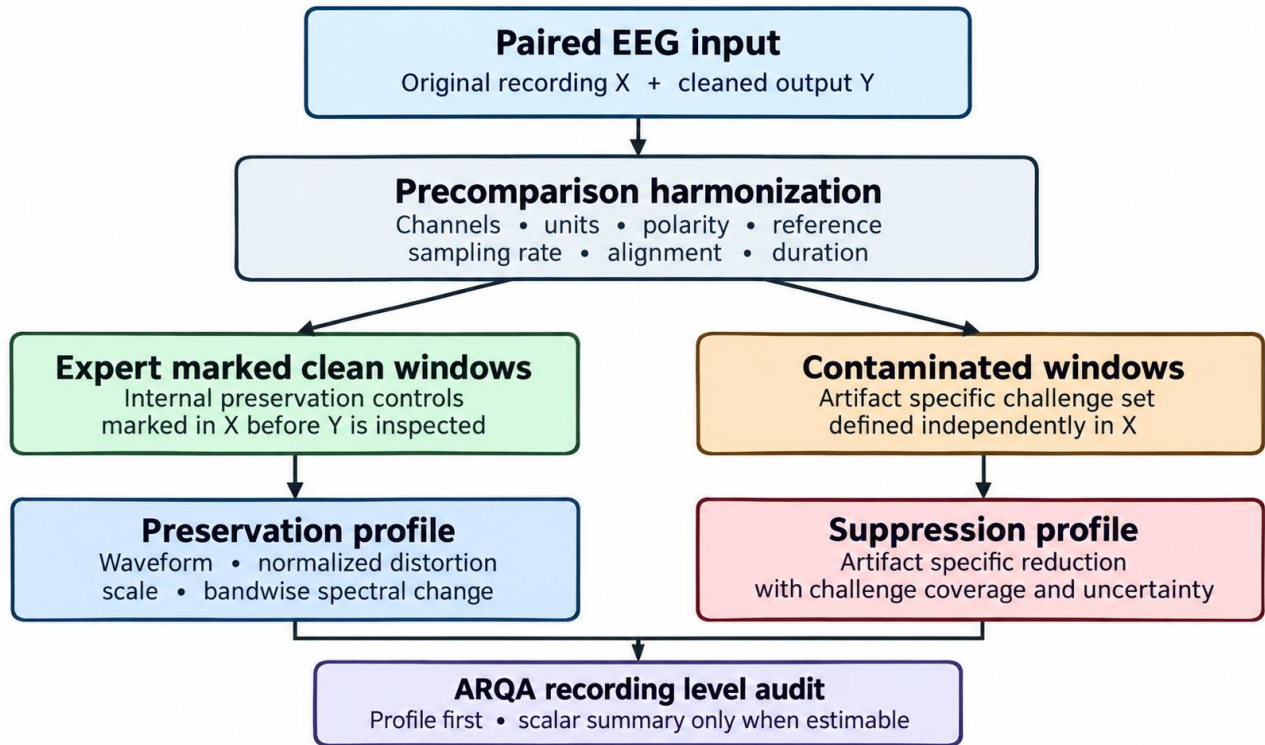


Figure 1. ARQA workflow. The original and cleaned signals are first expressed in a common comparison space. Expert-marked clean windows quantify preservation, whereas contaminated windows challenge artifact suppression. Coverage and uncertainty accompany both domains. A scalar summary is secondary and is not reported when either domain is insufficiently challenged.

2.2 Pre-comparison signal harmonization

Differences between X and Y can be caused by the cleaning method or by mundane mismatches introduced during import and export. ARQA therefore treats harmonization as a mandatory gate, not as an optional preprocessing detail. The following quantities must be reconciled before any quality metric is computed:

- channel identity and order, with a documented rule for dropped or interpolated channels;
- physical units and calibration, including microvolt-to-volt conversion;
- channel polarity and montage conventions;
- reference space, preferably by transforming both signals to the same explicitly named reference;
- sampling frequency and anti-aliasing procedure;
- sample-level temporal alignment and any filter or algorithmic delay; and
- common start time and duration after trimming unequal edges.

The need for this gate was demonstrated in Recording B. Its original EEG was referenced to A2, whereas the exported GEDAI signal used an average reference. Direct comparison produced spurious channel-specific distortions. Re-referencing the original 19-channel EEG to the common average removed this mismatch. A residual alignment check should then maximize or verify cross-correlation at or near zero lag in known clean intervals without permitting unconstrained warping of the data.

2.3 Preservation metrics in clean windows

ARQA reports complementary metrics because no single number distinguishes waveform shape, amplitude scale, and spectral content. Metrics are computed by channel and summarized across

channels with the mean, dispersion, and range; the channel-level distribution should remain available for inspection.

$$r_c = \text{corr}(x_c, y_c), \quad D_c = \frac{\text{Var}(x_c - y_c)}{\text{Var}(x_c)}, \quad q_c = \frac{\sigma(y_c)}{\sigma(x_c)}$$

Waveform correlation r_c measures similarity of shape but is largely invariant to positive amplitude rescaling. Normalized difference variance D_c , termed ENOVA_{clean} in the pilot analyses, quantifies the energy of the change relative to that of the original clean signal. The scale ratio q_c reveals systematic compression or amplification. Correlation must therefore not be described as amplitude-sensitive, and r_c and D_c must not be treated as independent. For centered signals, their exact relationship is:

$$D_c = 1 + q_c^2 - 2r_c q_c \quad \text{for centered signals}$$

This identity is diagnostically useful. A recording can have r_c near one and still have a non-negligible D_c if q_c differs from one. Conversely, the frequently used approximation $D_c \approx 1 - r_c^2$ holds only under additional assumptions and is not used here.

Spectral preservation is reported as the bandwise log-power change for each band b , computed on the same expert-marked clean samples:

$$\Delta P_b = 10 \log_{10} \left(\frac{P_{b,\text{after}}}{P_{b,\text{before}}} \right)$$

A value of 0 dB indicates preservation, negative values indicate attenuation, and positive values indicate amplification. Artifact-band reduction within clean controls does not measure efficacy; it reflects collateral spectral modification. Consequently, the earlier spectral selectivity index, which subtracted alpha change from delta change, is not used as a primary metric. A method should not receive credit merely because it attenuated an artifact-dominated band more strongly than it attenuated a neural band.

2.4 Artifact suppression metrics

Efficacy is evaluated only in windows classified as contaminated in X and must be tied to the artifact under study. Low-frequency amplitude or power can be appropriate for head-motion and ocular transients; high-frequency power or spectral slope can index muscle activity; narrow-band power can index line contamination; and peak amplitude or slope can characterize electrode pops. Whenever a dedicated EOG, ECG, accelerometer, impedance, or external reference is available, it should be preferred over a nonspecific EEG proxy.

A generic power-based suppression value may be expressed as $-\Delta P_a$, where ΔP_a is the change, in decibels, of an artifact marker within contaminated windows. This quantity is interpretable only if the marker is demonstrably dominated by the targeted artifact. ARQA does not infer the inaccessible clean neural waveform underlying a contaminated interval; therefore, strong suppression alone cannot establish successful reconstruction.

2.5 Coverage, confidence, and the status of a composite score

Every ARQA report includes the number and proportion of clean-control and artifact-challenge windows, their total duration, and their distribution across the recording. This information indicates whether preservation and efficacy were adequately evaluated. Confidence intervals can be estimated

using a block bootstrap over windows to preserve temporal dependence. A minimum number and temporal distribution of windows should be specified prospectively for each application.

The pilot implementation combined five quantities into a 0-100 Artifact Removal Index (ARI). This scalar is retained only as an exploratory development target. The initial weights were not empirically calibrated; correlation and normalized distortion are mathematically coupled; and an efficacy term can penalize a nearly clean recording simply because little artifact was available to remove. ARQA version 1 therefore treats the preservation, suppression, and coverage profiles as primary. A composite ARI should be reported only after its weights and decision thresholds have been validated in a larger labeled dataset, and it should be withheld whenever either control or challenge coverage is insufficient.

Table 1. Required components of an ARQA report.

Domain	Primary quantities	Interpretation	Required context
Preservation	r_{clean} ; D_{clean} ; scale ratio; bandwise ΔP	Collateral modification of expert-marked clean EEG	Number and duration of clean windows; channel distribution
Suppression	Artifact-specific amplitude or power reduction	Reduction of the targeted contaminant	Artifact type; marker validity; number of contaminated windows
Coverage	Clean fraction; challenge fraction; temporal distribution	How well each domain was tested	Prospective adequacy rule and uncertainty interval
Exploratory ARI	Predeclared combination calibrated	Optional compact summary	Withhold if unvalidated or insufficiently challenged

3 Proof of concept

3.1 Recordings and anonymization

Two de-identified, eyes-closed resting-state EEG recordings were used as contrasting examples. No participant names or clinical identifiers are included; the cases are designated Recording A and Recording B. Both recordings contained standard 10-20 EEG channels, and all comparisons used the 19 channels common to both the original and processed files. The raw clinical EEG recordings are not distributed with this report.

Table 2. Characteristics of the two de-identified proof-of-concept recordings.

Characteristic	Recording A	Recording B
Artifact profile	High-load generalized cranial movement artifacts	Low-load occasional transients
Duration	1,351 s (22.5 min)	4,141 s (69.0 min)
Original sampling rate	500 Hz	200 Hz
Comparison sampling rate	200 Hz after polyphase resampling	200 Hz
Original reference	FCz-referenced channel labels	A2 mastoid reference

Characteristic	Recording A	Recording B
Comparison reference	Common-average reference across 19 channels	Common-average reference across 19 channels
Original amplitude range	-2,011 to +2,082 μV	-239.7 to +271.0 μV
Post-GEDAI range	-28.8 to +30.7 μV	-113.5 to +97.8 μV
Expert-marked clean windows	241 of 675 (35.7%)	2,041 of 2,070 (98.6%)
Artifact-challenge windows	434 of 675 (64.3%)	29 of 2,070 (1.4%)
Post-GEDAI v1.7 range	-26.6 to +27.7 μV	-96.8 to +88.1 μV
v1.7 clean-window r_{clean} (mean)	0.718	0.987
v1.7 ENOVA ₁ clean (mean)	45.97%	2.52%
v1.7 α ΔP at O1 (clean windows)	-0.4 dB (-09%)	-0.1 dB (-2%)

3.2 Expert-marked clean controls and amplitude sensitivity

The primary preservation analysis used 2-s windows previously marked by an EEG specialist as suitable for qEEG. The markings were made on the original signals and fixed before the cleaned signals were evaluated. Peak-to-peak amplitude was retained as a secondary descriptor and sensitivity analysis rather than as the source of the primary labels. At a common descriptive threshold of 100 μV across the 19 channels, the dataset contained 241 low-amplitude windows in Recording A and 2,041 in Recording B, corresponding to clean-control coverages of 35.7% and 98.6%, respectively.

Because amplitude alone cannot identify all ocular, muscle, line-noise, or electrode artifacts, future validation should compare expert markings with artifact-specific detectors and report inter-rater agreement. The expert markings in this proof of concept establish an independent test condition but do not constitute a universally reproducible labeling rule.

3.3 GEDAI processing and analysis

Both recordings were processed with GEDAI v1.5 in EEGLAB using the same documented configuration: automatic mode, 12 epoch cycles, a 0.5-Hz low-frequency cutoff, the precomputed leadfield option, a parameter setting of 0.95, the EEG signal type, and otherwise the supplied infinite/default limits. The original signal from Recording A was resampled from 500 to 200 Hz. The common 19-channel signals were then expressed in the same reference space and aligned before the ARQA metrics were computed. To assess the sensitivity of ARQA to algorithm version, both recordings were additionally processed with GEDAI v1.7 using the identical configuration. The v1.7 output served as a supplementary test of whether ARQA could detect version-specific differences in the preservation-suppression profile within the same recording.

The pilot analyses were conducted in Python using NumPy 1.26 and SciPy 1.13. Pearson correlation was computed by channel on concatenated clean samples. Normalized difference variance was computed as $\text{Var}(X-Y)/\text{Var}(X)$. Band-limited amplitudes were obtained using fourth-order Butterworth filters applied forward and backward, and power spectra were estimated using Welch's method with 4-s segments. The results are descriptive because the study comprises two purposively selected recordings rather than a statistical sample.

4 Results

4.1 Preservation and evaluation coverage

The same cleaning method produced markedly different preservation profiles. In Recording A, the mean clean-window correlation was 0.444, and the mean normalized difference variance was 82%. These values indicate substantial modification of the signal, even within windows that had been judged suitable for qEEG before cleaning. The separate full-recording correlation was 0.134; this value is distinct from the clean-window correlation and is not used interchangeably with it.

In Recording B, the mean clean-window correlation was 0.9973 ± 0.002 , and the normalized difference variance was 0.64%. Across channels, clean-window correlations ranged from 0.9917 to 0.9994, and normalized distortion ranged from 0.14% to 1.67%. Bandwise changes within clean controls remained within ± 0.6 dB. These results are consistent with near-complete preservation of the already clean signal. When GEDAI v1.7 was applied to the same two recordings, ARQA revealed a version-specific asymmetry. In Recording A, v1.7 substantially improved clean-window preservation: mean r_{clean} rose from 0.134 to 0.718, and $\text{ENOVA}_{\text{clean}}$ fell from 82% to 46%. Alpha-band modification at O1 decreased from -10.1 dB to -0.4 dB, indicating that v1.7 largely preserved posterior neural oscillations in segments that v1.5 had substantially altered. In Recording B, preservation remained excellent under both versions (r_{clean} 0.9973 vs 0.9874; $\text{ENOVA}_{\text{clean}}$ 0.64% vs 2.52%), with the small regression concentrated at the frontal poles (FP1/FP2 ENOVA : 1.1% vs 7.7%). This version-specific asymmetry—v1.7 markedly better in the high-artifact recording, marginally worse in the clean recording at frontal electrodes—was detectable by ARQA but would not have been evident from the GEDAI internal diagnostics alone, since the SENSAT score was comparable between versions (72.6% for both).

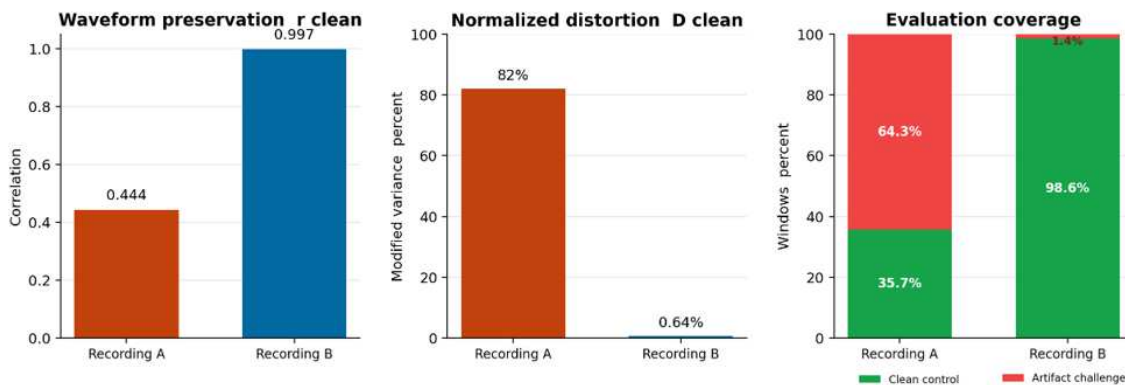


Figure 2. Contrasting recording-level ARQA profiles. Higher r_{clean} indicates better waveform preservation, whereas lower D_{clean} indicates less collateral modification. Recording A provides substantial artifact-challenge coverage but poor preservation. Recording B provides excellent preservation evidence but only 1.4% artifact-challenge coverage; therefore, efficacy is weakly identified.

Table 3. Primary ARQA results. The values distinguish between full-recording and clean-window analyses.

Measure	Recording A	Recording B	ARQA interpretation
Clean-window r_{clean}	0.444	0.9973 ± 0.002	A poor; B excellent
Full-recording r_{full}	0.134	Not required for primary preservation	Do not confuse global and clean-window r
D_{clean}	82%	0.64%	A severe modification; B minimal modification

Measure		Recording A	Recording B	ARQA interpretation
Clean-window spectral change		O1 alpha-amplitude reduction of 43%	All reported bands within ± 0.6 dB	A loses physiologic alpha even in controls
Full-recording delta-amplitude reduction		91.9% mean	Small overall change	A shows strong low-frequency attenuation
Full-recording alpha-amplitude reduction		77.5% mean	Small overall change	A suppression is not spectrally selective
Clean-control coverage		35.7% (241 windows)	98.6% (2,041 windows)	Preservation estimable in both
Artifact-challenge coverage		64.3% (434 windows)	1.4% (29 windows)	B efficacy insufficiently challenged

Table 4. ARQA results by recording and GEDAI version.

	Recording A		Recording B	
Measure	v1.5	v1.7	v1.5	v1.7
Clean-window r_{clean} (mean)	0.134	0.718	0.9973	0.987
ENOV r_{clean} (mean)	82%	46%	0.64%	2.52%
Alpha ΔP at O1 (clean windows)	-10.1 dB	-0.4 dB	-0.1 dB	-0.1 dB
FP1/FP2 ENOV r_{clean} (mean)	94%	92%	1.1%	7.7%
Clean-window fraction ($\theta=100\mu\text{V}$)	35.7%	67.3%	98.6%	98.6%
Post-GEDAI amplitude range	$\pm 31\mu\text{V}$	$\pm 27\mu\text{V}$	$\pm 114\mu\text{V}$	$\pm 97\mu\text{V}$
Artifact power reduction (APR)	~95%	96.6%	53%	59.1%
GEDAI internal SENS r_{AI} score	72.6%	72.6%	73%	73%

Green: excellent preservation; amber: marginal; red: poor. SENS r_{AI} is the algorithm-internal score (identical for both versions, confirming that ARQA adds information not captured by internal diagnostics). FP1/FP2 ENOVA in Recording B increased from 1.1% (v1.5) to 7.7% (v1.7), indicating a minor frontal regression in v1.7 on already-clean signal.

4.2 Artifact suppression

Recording A showed marked attenuation across the full signal: mean delta-band amplitude decreased by 91.9%, alpha-band amplitude by 77.5%, and beta-band amplitude by 75.9%. Delta-band power changed by an average of approximately -21.9 dB, but alpha-band power also changed by approximately -13.4 dB. The post-GEDAI amplitude range contracted from approximately $\pm 2,082$ μV to ± 31 μV . These observations demonstrate strong suppression; however, the parallel loss of neural-frequency activity and the poor clean-window preservation profile argue against interpreting the output as selective artifact removal.

Recording B contained only 29 artifact-challenge windows. The pilot estimate of artifact-power reduction was approximately 53%, but this estimate is unstable and difficult to interpret because only 1.4% of the windows challenged efficacy. ARQA therefore characterizes Recording B as showing excellent preservation but insufficient evidence regarding removal efficacy. It should not be summarized merely by a middling scalar score. Under GEDAI v1.7, artifact-power reduction in Recording A increased slightly to 96.6% (vs 95% for v1.5), and the clean-window fraction nearly doubled from 35.7% to 67.3%, indicating that v1.7 produced a less aggressively flattened output. In

Recording B, suppression increased marginally from 53% to 59.1%, but the 29-window challenge set remained too small for a stable efficacy estimate in either version.

4.3 Sensitivity to an amplitude descriptor

Amplitude thresholds were explored to characterize, rather than replace, the expert labels. Recording B remained predominantly low-amplitude across a wide range of thresholds, whereas Recording A displayed a much heavier high-amplitude tail. This marked threshold dependence confirms why the detector and its operating threshold must accompany every ARQA report.

Peak-to-peak threshold	Recording A low-amplitude windows	Recording B low-amplitude windows
50 μ V	—	1,501 of 2,070 (72.5%)
75 μ V	—	1,999 of 2,070 (96.6%)
100 μ V	241 of 675 (35.7%)	2,041 of 2,070 (98.6%)
150 μ V	—	2,057 of 2,070 (99.4%)
200 μ V	482 of 675 (71.4%)	2,066 of 2,070 (99.8%)

Table 4. Secondary threshold-response description. These counts do not replace the expert qEEG markings used as the primary preservation controls.

5 Discussion

5.1 What the proof of concept establishes

The two cases illustrate the practical value of a recording-level audit. GEDAI cannot be characterized globally as good or bad on the basis of these data. Under the low-artifact conditions of Recording B, the algorithm left expert-marked clean EEG essentially unchanged. Under the high-load generalized movement conditions of Recording A, it strongly attenuated the recording but also altered the clean controls and neural-frequency power. The actionable conclusion is conditional: the impact of a cleaning method depends on the structure of the individual recording, and that impact can be measured after processing.

This conclusion cannot be derived from the visual smoothness of Y. Nor is it guaranteed by an algorithm-internal score, because such a score answers a model-specific question. For GEDAI, leadfield-subspace separation is scientifically meaningful, but its internal diagnostics do not, by themselves, quantify preservation of the particular original recording. ARQA complements those diagnostics by measuring the paired output within an explicitly defined comparison space. The version comparison between GEDAI v1.5 and v1.7 provides a concrete illustration of this complementarity. Both versions produced nearly identical SENSAT scores (72.6%), yet ARQA revealed a substantial difference: v1.7 raised r_{clean} in Recording A from 0.134 to 0.718 and reduced alpha attenuation at O1 from -10.1 to -0.4 dB, while in Recording B it introduced a small frontal regression (FP1/FP2 ENOVA 1.1% to 7.7%) invisible from the internal score. This asymmetric pattern—better selectivity in high-artifact recordings, marginally higher frontal collateral modification in clean ones—is precisely the kind of version-specific trade-off that a recording-level audit is designed to detect and that internal algorithm diagnostics cannot resolve by themselves.

5.2 Relationship to prior work and defensible novelty

ARQA belongs to a growing family of evaluation strategies that jointly consider artifact reduction and signal preservation. RELAX and RELAX-Jr use complementary signal-to-error and artifact-to-residue metrics and have shown that pipelines may occupy different preservation-suppression trade-offs (Bailey et al., 2023a; Hill et al., 2024). Bailey et al. (2025) further demonstrated the advantage of targeting cleaning to the periods and frequencies in which artifacts occur. Nyanney et al. (2026) evaluated detection-guided methods against expert-annotated clean windows using familiar fidelity measures.

Against this background, the defensible novelty of ARQA is not the discovery that clean data should remain clean. Rather, it is the formalization of that expectation as an algorithm-independent, post hoc audit of arbitrary paired EEG files at the level of an individual recording; the mandatory harmonization gate; the explicit separation of preservation, suppression, and evaluation coverage; and its intended use as a quality-control step before qEEG analysis. This positioning avoids an unsustainable priority claim while retaining a distinct practical contribution.

5.3 Why a profile is preferable to an uncalibrated rank

The pilot ARI values of 28 for Recording A and 70 for Recording B were initially useful for development, but they expose the limitations of a single weighted score. Recording B preserved clean EEG almost perfectly, yet its score was reduced because little artifact was present to remove. A scalar also obscures the distinction between waveform correlation and amplitude scaling and may double-count coupled quantities. Until weights and decision thresholds are derived from external outcomes, ARQA should report a profile. If a future scalar is required for automated gating, it should be calibrated in an independent labeled dataset, include uncertainty, and apply explicit insufficiency rules rather than converting an absence of challenge into failure.

5.4 Implications for qEEG

For qEEG, preservation must be assessed using the same variables that will subsequently support inference. A cleaning procedure that substantially attenuates posterior alpha activity can shift absolute power, spectral ratios, peak-frequency estimates, and age-normed z scores, even if movement amplitude is reduced. The Recording A profile therefore does not support the use of the cleaned output for normative spectral interpretation without reprocessing. Recording B provides strong evidence that its clean portions were preserved, but its few contaminated windows still require visual and artifact-specific review. ARQA is a quality-control layer; it does not replace expert inspection, raw-data retention, or transparent reporting of the preprocessing pipeline.

5.5 Limitations

This study has several deliberate limitations. First, two selected recordings constitute a proof of concept rather than a validation sample, and no inferential claim about GEDAI or other methods is justified. Second, the expert markings were provided as established qEEG windows; inter-rater reliability and blinding beyond the fixed pre-output marking procedure were not evaluated. Third, contaminated windows do not provide ground truth for the latent neural signal, so efficacy must be assessed using artifact-specific proxies. Fourth, Recordings A and B differ in duration, original sampling rate, reference, and artifact load; harmonization prevents known technical mismatches but cannot make the cases exchangeable. Fifth, the current demonstration evaluates a single method and

does not test ICA, artifact subspace reconstruction, wavelet cleaning, or epoch rejection. Finally, the proposed metrics and any future ARI thresholds require calibration against simulations, external sensors, expert ratings, and the stability of downstream qEEG measures.

5.6 A question readers may reasonably ask

A natural question arises from the present proof of concept: if an expert has already reviewed the entire recording and identified the segments suitable for quantitative EEG analysis, why is an additional framework such as ARQA needed?

In the present study, expert-marked clean EEG segments were deliberately used as fixed internal controls to test the conceptual and quantitative foundations of ARQA independently of the performance of an automated detector. This design allows the effect of the cleaning procedure itself to be evaluated without introducing uncertainty from an additional signal-selection algorithm. Manual marking, however, is not intended to be a required step in the operational implementation of ARQA.

The planned operational implementation of ARQA includes an automatic clean-signal detection module designed to identify, throughout an EEG recording, the segments suitable for quantitative analysis. Its purpose is therefore broader than evaluating whether an artifact-removal algorithm altered previously identified clean EEG. The automatic module is intended both to provide the internal preservation controls required by ARQA and to identify usable EEG directly from the original recording, thereby reducing or potentially eliminating the need for exhaustive manual review in routine applications.

Operational architecture of ARQA

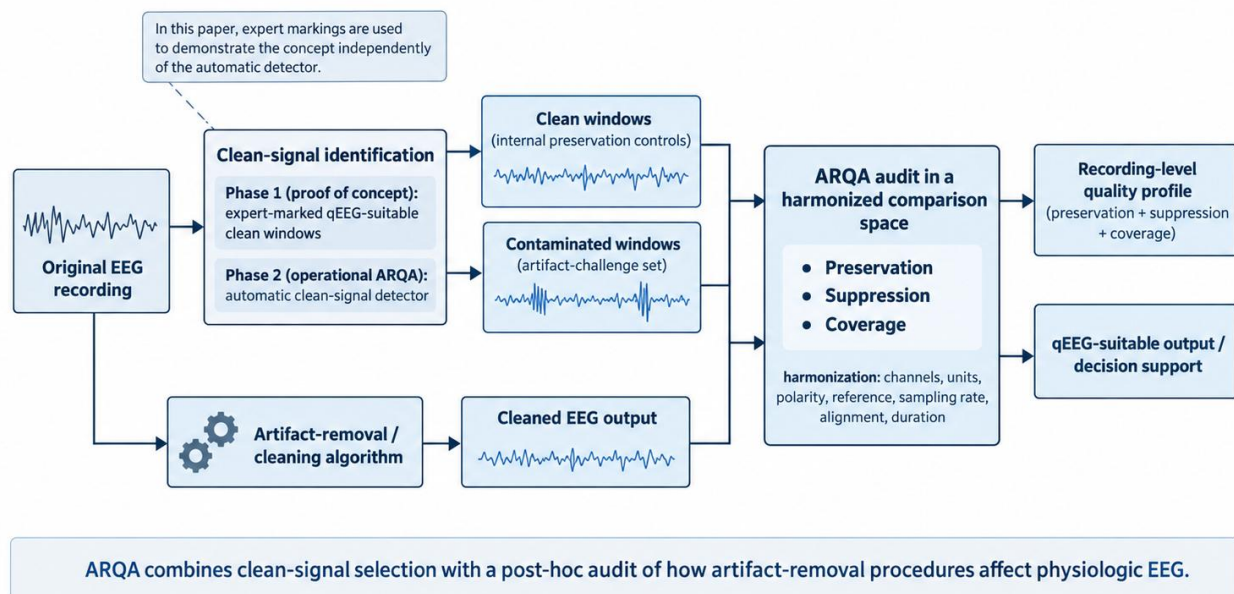


Figure 3. Proposed operational architecture of ARQA. The present proof of concept uses expert-defined clean controls. Phase 2 will validate the automatic detector that supplies qEEG-suitable clean controls and candidate artifact-challenge intervals. The original and processed signals pass through a mandatory harmonization gate before preservation, suppression, coverage, and uncertainty are evaluated. The unified workflow produces both automatically selected qEEG-suitable segments and an auditable recording-level processing profile.

This distinction is important. In the present proof of concept, expert annotation provides the reference condition used to demonstrate the ARQA framework. In the next phase, clean-window identification will become an automated component of the framework. The resulting system is intended to answer two related questions: (1) which portions of the original recording are sufficiently artifact-free for qEEG analysis, and (2) whether a subsequent artifact-removal procedure preserves this physiological signal while selectively reducing contamination elsewhere.

An automatic clean-signal detector has already been implemented and is currently undergoing testing. It is not presented as a validated tool in this report because the required independent validation has not yet been completed. The planned Phase 2 validation will include evaluation in independent international EEG datasets, comparison with annotations from experienced clinical neurophysiologists, and cross-validation in a distributed multi-rater annotation study. Performance will be quantified in terms of agreement with expert-defined qEEG-suitable segments, robustness across recording conditions and laboratories, and generalization to datasets not used during algorithm development.

Once validated, this component should make ARQA substantially more scalable. Within a single workflow, the framework would then provide both an automatic criterion for identifying EEG suitable for quantitative analysis and an independent recording-level audit of the changes introduced by artifact-removal procedures. The present study therefore evaluates the auditing principle using expert-defined controls, whereas Phase 2 will address the independent validation and deployment of the automated clean-signal selection component.

6 Recommended development and validation program

The next phase of ARQA should move beyond the present proof of concept and develop the framework into a validated, scalable quality-control system for quantitative EEG. This program has two complementary objectives: first, to validate the recording-level audit of artifact-removal procedures introduced here; and second, to validate the automatic identification of EEG segments suitable for quantitative analysis, thereby reducing the need for exhaustive expert marking in routine use.

The development program can be organized around four interconnected aims.

Aim 1. Validate the automatic identification of qEEG-suitable clean signal

The first aim is to validate the automatic clean-signal detection module that will provide the internal preservation controls required by operational ARQA. The detector has already been implemented and is currently undergoing testing, but it requires independent validation before it can be considered a reliable marking tool.

Validation should use heterogeneous EEG datasets acquired in different laboratories, populations, recording systems, sampling rates, reference schemes, and artifact conditions. The reference standard should be based on annotations from experienced clinical neurophysiologists who apply prospectively defined criteria for qEEG-suitable signal.

Because expert annotation is not perfectly reproducible, the validation should explicitly quantify inter-rater agreement. A distributed multi-rater annotation study involving neurophysiologists from different centers would provide both a consensus reference and an estimate of the intrinsic

uncertainty of human labeling. The automatic detector should then be evaluated against individual raters and consensus labels using sensitivity, specificity, predictive values, balanced accuracy, agreement coefficients, and window- and recording-level performance measures.

A critical component of this aim will be external cross-validation. Datasets used for algorithm development should be strictly separated from those used for final evaluation, and at least part of the validation should be performed in independent international datasets that were not used for algorithm training or parameter optimization.

Aim 2. Validate ARQA across artifact-removal methods and artifact families

The second aim is to establish whether the preservation-suppression-coverage framework behaves consistently across substantially different artifact-removal strategies. The same original recordings should therefore be processed using representative methods, including GEDAI, ICA-based component subtraction, artifact subspace reconstruction, targeted wavelet or Wiener filtering, and conventional epoch rejection.

Evaluation should be stratified by artifact family rather than treating contamination as a single phenomenon. At a minimum, ocular activity, cranial and facial muscle activity, gross movement, cardiac contamination, line interference, unstable electrodes, and transient electrode artifacts should be represented.

For each recording and cleaning method, ARQA should quantify three distinct domains: preservation of EEG that did not require correction, suppression of the targeted artifact, and the amount and distribution of signal available to estimate each domain. This approach will allow cleaning procedures to be characterized by recording-specific quality profiles rather than by a single global ranking.

Aim 3. Establish ground truth and clinically relevant validation targets

Real contaminated EEG does not provide access to the unknown neural signal that would have been recorded in the absence of the artifact. Validation should therefore combine complementary sources of ground truth.

Synthetic and semi-synthetic datasets should be generated by adding realistic ocular, muscle, movement, cardiac, line-noise, and electrode artifacts to known clean EEG. These datasets will provide direct reconstruction ground truth and permit quantitative assessment of the neural signal removed or distorted by each cleaning method.

In real recordings, external measurements such as EOG, ECG, accelerometry, impedance information, or other artifact-specific sensors should be used whenever available. Expert consensus should provide an additional clinical reference.

Importantly, validation should not stop at signal-level similarity. The downstream stability of the variables for which the EEG was ultimately acquired should also be assessed. These variables should include absolute and relative spectral power, spectral ratios, peak alpha frequency, normative qEEG z scores, functional connectivity measures, and EEG source estimates. The framework should flag any cleaning procedure that improves visual appearance while materially altering these downstream quantities.

Aim 4. Develop a validated decision-support and quality-control system

The final aim is to integrate the automatic detector and the ARQA audit into a unified recording-level quality-control workflow.

The operational system would first identify the portions of the original EEG that are suitable for quantitative analysis. These windows would provide internal negative controls for evaluating any subsequent cleaning procedure, whereas contaminated intervals would provide artifact-specific challenge sets. The original and cleaned signals would then be transformed into a harmonized comparison space before preservation, suppression, and coverage are quantified.

In operational use, this first step would also support a practical decision that precedes artifact removal. If ARQA identifies a sufficient amount and temporal distribution of qEEG-suitable clean signal, the specialist may determine that the recording can be analyzed without applying a cleaning algorithm. If the available clean signal is insufficient, the same detection stage can be used to select the least contaminated candidate intervals for targeted cleaning. Cleaning would then be applied only where it is needed, reducing the risk of estimating algorithm parameters from severely contaminated segments and avoiding unnecessary modification of signal that has already been marked as clean.

The primary output should remain a multidimensional ARQA profile. A composite Artifact Removal Index should be introduced only if its weights and thresholds can be derived and validated against independent ground truth or clinically meaningful downstream criteria. Importantly, insufficient preservation or artifact-challenge coverage should produce an explicit 'insufficient evidence' state rather than an artificial numerical score.

Uncertainty should be quantified at the recording level using a block bootstrap or an equivalent procedure that preserves temporal dependence. Prospective adequacy criteria should define the minimum amount and temporal distribution of clean and contaminated data required for reliable estimation.

Multicenter validation and expected outcome

The definitive validation stage should use a multicenter design incorporating EEG recordings from different laboratories and acquisition systems. This design is essential because an operational qEEG quality-control system must generalize across differences in hardware, reference, sampling frequency, population, recording environment, and artifact prevalence.

The final product of this development program would be an independently validated, method-agnostic framework capable of automatically identifying qEEG-suitable signal and auditing how artifact-removal procedures modify an individual EEG recording. Such a system could provide a standardized quality-control layer between EEG acquisition and quantitative analysis while preserving the original data and maintaining explicit traceability of every processing decision.

The present study constitutes Phase 1 of this program: demonstration of the recording-level audit using expert-defined internal controls. Phase 2 will validate automatic clean-signal identification and inter-rater reference standards. Phase 3 will conduct cross-method, cross-artifact, and multicenter validation, including the evaluation of downstream qEEG outcomes. Phase 4 will integrate these components into a distributable ARQA software implementation and establish criteria for prospective clinical and research use.

7 Conclusions

ARQA is a method-agnostic, recording-level audit of the changes introduced by an EEG artifact-removal procedure. Expert-marked clean segments of the original recording serve as internal preservation controls, whereas contaminated segments provide an artifact-specific challenge set. The framework requires signal harmonization and reports preservation, suppression, and coverage separately. In the present proof of concept, the same GEDAI v1.5 configuration produced severe collateral modification in Recording A and near-complete preservation in Recording B, although Recording B contained too little contaminated material to support a strong conclusion about efficacy. Applying GEDAI v1.7 to the same recordings revealed a version-specific trade-off that was not reflected in the identical SENSAI scores: substantially better preservation in the high-artifact recording, with a small frontal regression in the clean recording. ARQA detected this difference; the internal diagnostic did not. This is precisely the information that a quality-control framework should reveal. ARQA should therefore be developed first as a transparent profile; any composite index should remain secondary until it has been empirically calibrated.

Ethics approval and consent to participate

This proof-of-concept is a retrospective secondary methodological analysis of two EEG recordings acquired during routine clinical care at CIMMCO (Centro Integrador del Movimiento, Mente y Conducta), Querétaro, Mexico. The study was conducted in accordance with the principles of the Declaration of Helsinki, using de-identified retrospective clinical EEG data. Both patients provided written informed consent for the clinical procedures, scientific use and publication, and for the processing of their personal and health data under CIMMCO's informed consent and privacy procedures. The recordings are reported exclusively as Recording A and Recording B, and remain stored on a private computer with restricted access.

Consent for publication

The written informed consent obtained at CIMMCO includes authorization to use de-identified clinical information in scientific publications. This manuscript contains no names, dates of birth, facial images, or other directly identifying information. The two recordings are referred to exclusively as Recording A and Recording B.

Data availability

The raw EEG recordings analyzed in this proof of concept originated from clinical evaluations performed at CIMMCO and are not publicly available because they contain confidential clinical data. The source recordings are stored on a private computer with restricted access. The manuscript reports the de-identified recording characteristics, processing parameters, derived signal-quality measures, and aggregate results needed to evaluate the methodological conclusions. The ARQA framework is independent of these particular recordings and can be applied to any paired original and processed EEG datasets that satisfy the required signal-harmonization conditions.

Code availability

The reference implementation of ARQA is under active development. The mathematical definitions, signal-harmonization requirements, preservation metrics, suppression metrics, and processing

sequence required to reproduce the analyses presented here are described in the Methods. The automatic clean-signal detection component planned for Phase 2 has been implemented and is currently undergoing testing. Because independent cross-validation has not yet been completed, it is not released as a validated EEG-marking tool with this proof of concept. A public reference implementation is planned after completion of the validation phase.

Competing interests

The authors declare no competing financial or non-financial interests related to this work.

CRedit authorship contribution statement

Jorge F. Bosch-Bayard: Conceptualization; methodology; software; formal analysis; data curation; visualization; writing - original draft; writing - review and editing; project administration; supervision. **Lilia Morales Chacón, Alfonso García, Lourdes Cubero, Rubén Pérez Elvira, and Giuseppe A. Chiarenza:** Contributed equally to expert visual inspection of the EEG recordings, marking of EEG windows suitable for quantitative EEG analysis, evaluation of the EEG after artifact-removal procedures, and critical review and editing of the manuscript. **Judith Guerrero-Sauzamedia:** EEG data acquisition; organization and storage of the clinical EEG data; manuscript writing; review and editing of the manuscript. **Lidice Galan-Garcia, and Muhammad Usman Mustafa:** Methodology; manuscript review and editing. **Faisal Mushtaq, Jochem Rieger, Tomas Ros, Alan C. Evans, and Pedro A. Valdes-Sosa:** Conceptualization; contribution to methodological development; manuscript review and editing; supervision, project organization and scientific oversight.

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